

Comparative Assessment: LTZGLUE vs. GLUE

Deployment context: Luxembourg Public Administration — Civil Servant NLU Deployment

Deployment Context

Use case: In a public administration setting, a civil servant uses LLM-powered software to manage Luxembourgish citizen feedback by applying a model evaluated on topic classification, named entity recognition, intent detection, and sentiment analysis. The system automatically categorizes messages by domain and identifies key entities while classifying user intent and gauging sentiment to help detect frustrated users. These outputs enable the automated routing of requests to specific departments and the immediate prioritization of disgruntled or urgent correspondence, ensuring administrative actions are triggered accurately based on the content and tone of the message.

Target population: Civil servants in Luxembourgish government agencies who process digital correspondence and citizen inquiries written in the diverse orthographic styles of the Luxembourgish language.

Overall Risk Ratings

Benchmark	Risk	Avg. Score
LTZGLUE	HIGH	2.2 / 5
GLUE	HIGH	1.2 / 5

Dimension Scores Comparison

Dimension	LTZGLUE	Rating	GLUE	Rating	Δ
Input Ontology (IO)	2/5	Significant gaps	1/5	Serious concern	+1
Input Content (IC)	2/5	Significant gaps	1/5	Serious concern	+1
Input Form (IF)	3/5	Moderate gaps	1/5	Serious concern	+2
Output Ontology (OO)	1/5	Serious concern	1/5	Serious concern	0
Output Content (OC)	2/5	Significant gaps	1/5	Serious concern	+1
Output Form (OF)	3/5	Moderate gaps	2/5	Significant gaps	+1
Average	2.2		1.2		+1

Δ = LTZGLUE score – GLUE score. Positive = regional benchmark scores higher.

Per-Dimension Justification Comparison

Brief excerpts from each benchmark's assessment justification. Full justifications are available in the individual dimension PDFs.

Input Ontology (IO)

Whether the benchmark's test case categories cover the query types expected in deployment.

LTZGLUE — 2/5 (Significant gaps) [high confidence]	GLUE — 1/5 (Serious concern) [high confidence]
LTZGLUE provides a GLUE-style suite of eight tasks covering binary and multi-class sentence- and token-level classification [Q11, Q67], which is a credible general-purpose NLU foundation for Luxembourgish. However, for the specific deployment — routing citizen correspondence to government departments — the taxonomies are misaligned.	GLUE's task ontology consists of nine English sentence-understanding tasks — sentiment, paraphrase, entailment, QA, acceptability, similarity [Q3, Q18] — with no administrative-domain categories. The deployment requires routing categories that GLUE does not contain: cross-border worker (frontalier) classification, national vs. communal (État vs.

Input Content (IC)

Whether individual datapoint content is culturally and contextually appropriate for the target region.

LTZGLUE — 2/5 (Significant gaps) [high confidence]	GLUE — 1/5 (Serious concern) [high confidence]
Benchmark inputs are drawn predominantly from RTL news articles, the Luxembourgish Online Dictionary, Wikipedia/Wikidata, and a translated English voice-assistant corpus [Q12, Q16, Q24, Q30, Q34, Q46, Q51].	GLUE's input content is entirely English, drawn from American/British movie reviews [Q26], US/UK news [Q29, Q48], Wikipedia [Q41], Quora [Q31], linguistics journals [Q21], and English-language fiction and transcribed speech [Q38].

Input Form (IF)

Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

LTZGLUE — 3/5 (Moderate gaps) [medium confidence]	GLUE — 1/5 (Serious concern) [high confidence]
Form mismatch is moderate. Both benchmark and deployment are text-only in Latin script, so there is no modality or script-direction violation [regional context infrastructure_notes.deployment_modality; writing_systems].	GLUE inputs are standardized written English text in a single script with consistent orthography, formatted as single sentences or sentence pairs [Q4, Q15]. The deployment input is text in three languages with frequent code-switching at lexical, phrasal, and sentence level [WEB-11], non-standardized Luxembourgish spelling where 'large amount of ort ...

Output Ontology (OO)

Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

LTZGLUE — 1/5 (Serious concern) [high confidence]	GLUE — 1/5 (Serious concern) [high confidence]
Severe structural misalignment between benchmark output ontology and deployment requirements. All eight LTZGLUE tasks produce single hard labels [Q67] across narrow label spaces: binary headline acceptability [Q148], three-class sentiment [Q149], binary/four-class linguistic acceptability [Q33, Q151], BIO-tagged NER over PER/ORG/LOC/MISC/DATE [Q44, ...	GLUE's output ontology is hard single-label classification (binary or three-class) plus one regression task [Q12], scored via accuracy/MCC/F1/Pearson [Q20, Q23, Q30, Q33, Q35] aggregated as a macro-average [Q59, Q60].

Output Content (OC)

Whether ground-truth labels align with the judgments of regional stakeholders.

LTZGLUE — 2/5 (Significant gaps) [high confidence]	GLUE — 1/5 (Serious concern) [high confidence]
Annotation provenance and quality are mixed and weakly tied to civil-administration norms. Sentiment was annotated by two native LTZ speakers with Cohen's Kappa 0.45 — moderate agreement — and disagreements resolved by consensus [Q25, Q26, Q27, Q28]; no documentation of administrative-register expertise.	Annotator demographics for the main inherited GLUE tasks are not documented in the paper [Q6 establishes the inherited-dataset reliance; no demographic disclosure for SST-2 [Q26] or MNLI crowdworkers [Q36]].

Output Form (OF)

Whether the expected output modality matches regional deployment needs and accessibility.

LTZGLUE — 3/5 (Moderate gaps) [medium confidence]	GLUE — 2/5 (Significant gaps) [high confidence]
Output form is text-classification labels evaluated by macro-F1 with results averaged over three runs and standard deviations reported [Q93, Q96]. This is a standard, well-documented evaluation methodology and matches the deployment's text-to-label format at surface level.	GLUE evaluates text-based outputs only via hard single-label scoring on private test sets [Q57, Q58], with per-task scores aggregated as a macro-average [Q59, Q60] subject to two-submissions-per-day rate limiting [Q131].

Overall Summaries

LTZGLUE (risk: HIGH)

LTZGLUE is a credible first GLUE-style NLU benchmark for Luxembourgish and a valuable resource for general LTZ NLP capability assessment. However, for the specific deployment of routing citizen correspondence in Luxembourg public administration, validity is materially limited across the four HIGH-priority dimensions (IO, IC, OO, OC).

GLUE (risk: HIGH)

GLUE is fundamentally invalid as an evaluation benchmark for a Luxembourgish public-administration citizen-correspondence routing system. Five of six dimensions score 1 (major validity violations), with OF scoring 2 only because the surface text-in/label-out modality matches. The mismatches are total along every HIGH-priority dimension: (IO) GLUE's nine English NLU tasks contain zero administrative-routing categories and no national-vs-communal or frontier-vs-resident classifications [Q3, Q18, WEB-16, WEB-19]; (IC) all 409 sampled examples are English with US-political/cultural dominance, wh ...